

Mark schemes

Q1.

$$\frac{dv}{dt} = \frac{k}{v}$$

B1

$$\int v dv = \int k dt$$

Separation of variables involving t

M1

$$\frac{v^2}{2} = kt + c$$

m1

Integrate

A1

$$t = 0, v = u, \therefore c = \frac{u^2}{2}$$

m1

$$v^2 = u^2 + 2kt$$

A1

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[6]

Q2.

(a) $\mathbf{a} = 8 \cos t \mathbf{i} - 4 \sin t \mathbf{j}$

M1 differentiation

M1A1

$$\mathbf{F} = 2 \cos t \mathbf{i} - \sin t \mathbf{j}$$

A1

3

(b) (i) $|F| = \sqrt{4 \cos^2 t + \sin^2 t}$

Magnitude

M1

$$= \sqrt{3 \cos^2 t + 1}$$

CAO

m1

A1

3

(ii) $|\leq F| \leq 4$

B1 each value (at end of range)

B1B1

2

[8]

Q3.

(a) $R = 2 \times 9.8, \quad F = \frac{1}{7} \times 2 \times 9.8 = 2.8\text{N}$

*CAO M1: full method with 2×9.8 ,
accept inequality*

M1A1

2

(b) (i) $P = 1, \quad F = 1\text{N}$

B1

1

(ii) $P = 5, \quad F = 2.8$

Used

B1

$$5 - 2.8 = 2a$$

M1: 3 terms, with forces subtracted

M1A1

$$a = 1.1 \text{ ms}^{-2}$$

ft one slip

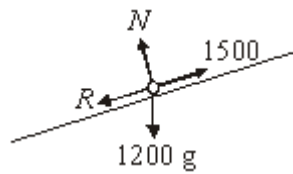
A1F

4

[7]

Q4.

(a) (i)



3 correct & labelled

B1

All correct & labelled, no extras

B1

2

Ignore additional components of weight

(ii) $1500 = R + 1200 \times 9.8 \times \frac{1}{14}$
M1: 3 terms with component attempted
($\alpha = 4.096$) – 1 each error

M1A2

$R = 660$
AWRT

A1

4

(iii) 15 ms^{-1}

B1

1

(b) (i) $\frac{1500}{1200}$
Accept $\pm$

M1

Mag of retardation = 1.25 ms^{-2}

A1

2

(ii) $0 = 15 + (-1.25) t$
velocities correct, but accept $\pm$ acceleration

M1

$t = 27.3 \text{ sec}$ (27.27)
ft one slip

A1F

Q5.

(a) $5mg - T = 5m \frac{g}{4}$

3 terms, recognisable, accept m missing or g missing once

M1

all correct & algebraic

A1

$T = 15 m \frac{g}{4} = 3.75 mg$

ft 9.8 used, one g missing, m missing, sign error provided T positive

A1F

3

(b) $T - kmg = km \frac{g}{4}$
as in (a)

M1A1

$k = 3$

CAO
following fully correct work in (a) & (b)

A1

3

(c) Force = $2T = \frac{15mg}{2} = 7.5 mg$

B1F

1

(d) $t = \frac{2}{3}$

$x = 0 + \frac{1}{2} \times \frac{g}{4} \times \frac{4}{9}$ accept $\left(\frac{2}{3}\right)^2$

method for x , and $\frac{g}{4}$ used for acceleration

M1

accept 9.8 substituted

A1

$$d = 2x = \frac{g}{9}$$

Fully correct, in terms of g

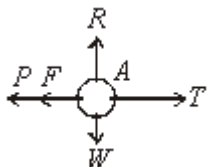
A1

3

[10]

Q6.

(a) (i)



For any 3 forces correct

B1

For the rest correct, and no extras
(accept F in either direction, but not μR)

B1

2

(ii) $R = 0.2 \times g$

$$F = 0.5 \times R$$

For equations for both R and F

M1

$$F = 0.98\text{N}$$

AG

A1

2

(iii) $T = 0.3 \times 9.8 = 2.94\text{N}$

M1A1

2

(iv) $P + F = T$

M1

least $P = 1.96\text{N}$

ft T provided $P > 0$

A1F

2

(b) (i) $2.94 - T = 0.3a$

M1A1

$T - 0.98 = 0.2a$

M1 either equation

(whole string method: 'correct' equation, M1 A1, answer A1; max 3 / 5)

A1

$1.96 = 0.5a$

m1

$a = 3.92\text{ms}^{-2}$

ft one error

A1F

5

(ii) $0.1 = \frac{1}{2} \times 3.92 \times t^2$

Use of candidate's acceleration, if 9.8

M1

$t = 0.226$ (0.22588)

ft candidate's acceleration (not 9.8) if

$a = 1.96$ used, $t = 0.319$

A1F

2

[15]

Q7.

Answers in (a), (b) and (c)(i) must be vectors

(a) $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$

Must be vector, both i and j present

B1

1

(b) $\mathbf{v} = 2t^2\mathbf{i} + 3t\mathbf{j}$

Integration of acceleration attempted

M1

ft a, both i and j present

A1F

2

(c) (i) $\mathbf{r} = \int_0^t (2t^2\mathbf{i} + 3t\mathbf{j}) dt$

Integration attempted

M1

$$= \frac{2t^3}{3}\mathbf{i} + \frac{3t^2}{2}\mathbf{j}$$

A1 each term. Condone i and j missing.

–1 for constants present,

–1 for both terms unsimplified

A1A1

3

(ii) $\left. \begin{array}{l} \mathbf{v} \text{ parallel to } \mathbf{i} + \mathbf{j} \\ 2t^2 = 3t \end{array} \right\}$

M1

$$t = \frac{3}{2}$$

Solve, ft v with vector components

A1F

$$\mathbf{r} = \frac{9}{4}\mathbf{i} + \frac{27}{8}\mathbf{j}$$

A1F

Distance $= \sqrt{\left(\frac{9}{4}\right)^2 + \left(\frac{27}{8}\right)^2}$

Magnitude of r in any form

M1

$$= 4.06 \text{ (4.056)}$$

Numerical answer, ft candidate's r and t

A1F

5

[11]

Q8.

(a) $mg - 0.1m v = m \frac{dv}{dt}$
Newton's 2nd law

M1

$$\frac{dv}{dt} = g - 0.1v$$

CAO

A1

2

(b) $\int_0^v \frac{dv}{g - 0.1v} = \int_0^t dt$
Attempt at integration with correct separation of variables

M1

$$[-10 \ln(g - 0.1v)]_0^v = [t]_0^t$$

-1 EE

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$$\ln \frac{g - 0.1v}{g} = -0.1t$$

A1F

$$v = 10g(1 - e^{-0.1t})$$

Or equivalent

A1F

5

(c) As $t \rightarrow \infty$, $e^{-0.1t} \rightarrow 0$

M1

1

$$\therefore v \rightarrow 10g = 98$$

[8]

Q9.

(a) $v^2 = u^2 + 2as$

Use of full method

M1

$$0^2 = 3.5^2 + 2 \times a \times 2.5$$

Correct subs

A1

$$a = -2.45 \text{ mag } 2.45$$

Magnitude required

A1

3

(b) (i) Friction : $0.2 \times 2.45 = 0.49\text{N}$

M1: use of $F = ma$ with ± 2.45

A1: Magnitude required

M1A1

2

(ii) $R = 0.2 \times g$

$$F = \mu R \quad 0.49 = \mu \times 0.2g$$

Use of $F = \mu R$ with $R = mg$ substituted

M1

$$\mu = 0.25$$

f.t. (i) provided μ positive

Use of $F < \mu R$, M1 A0

A1F

2

[7]

Q10.

(a) $1200 - R = 1000 \times 0.25$

M1: all relevant terms used

M1A1

$$R = 950 \text{ N}$$

f.t. one slip if $R > 0$

A1F

3

(b) $2100 - 1000g \times 0.1 - 950 = 1000 \times a$

M1: 3 or 4 terms considered

M1

*−1 each term incorrect (any error),
or missing, $\alpha = 5.74^\circ$*

A1A1

$a = 0.17 \text{ ms}^{-2}$

f.t. one error if 4 terms considered

A1F

4

[7]

Q11.

(a) (i) $T = 0.6a$

Either equation (M1 A0 for use of $0.1ga$)

M1

$0.1g - T = 0.1a$

SC who;e strng method:

A1

$0.1g = 0.7a$ M1A1 (total mass used)

A1

$0.1g = 0.7a$

$a = 1.4$ A1, max3/5)

m1

$a = 1.4 \text{ ms}^{-2}$

A1

5

(ii) $T = 0.6 \times 1.4 = 0.84 \text{ N}$

Dependent on M1 gained in (a),

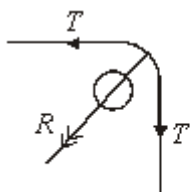
*Or, s.c. can gain M1 (from (a)) A1 here if
equations involving T not found in (a)*

Max M1 A1

A1

1

(iii)



$$R = 2T \cos 45^\circ$$

M1

recognising 2 tensions involved

B1

M1: attempt at Pythagoras or at a component of T

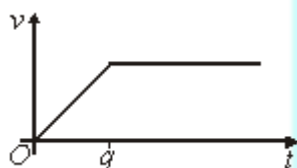
$$= 1.19 \text{ N}$$

A1: f.t. tension

A1F

3

(b)



1st line sloping and through O

B1

2nd line horizontal

B1

label at t q

B1

3

[12]

Q12.

(a) (i) $B: T = mg = 0.1 \times 9.8 = 0.98 \text{ N}$
cao

B1

1

(ii) $A: F = T (= 0.98) \quad R = 0.5 \times 9.8$

B1

$$\mu \times 0.5 \times 9.8 = 0.98$$

Condone inequality

M1

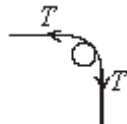
$$\mu = 0.2$$

cao

A1

3

(iii)



Must label T or 0.98 N

B1

1

(iv)



$$R = \sqrt{T^2 + T^2} \text{ or } 2T \cos 45^\circ$$

M1

$$R = 1.39\text{N}$$

A1

2

(b) (i) $A: T - 0.98 = 0.5a$

Either equation for M1

M1A1

$$B: 2 \times 0.1 \times 9.8 - T = 0.2a$$

A1

$$0.98 = 0.7a$$

m1

$$a = 1.4$$

cao

A1

5

(ii) $T = 0.98 + 0.5 \times 1.4$, $T = 1.68$
ft accuracy error in equation

A1F

1

(iii) $s = ut + \frac{1}{2} at^2$

$$0.7 = 0 + \frac{1}{2} \times 1.4t^2$$

M1

$$t = 1$$

A1

2

[15]

Q13.

(a) (i) $T = mg$
Accept $g = 9.8$ (or 10) used throughout (a)

B1

1

(ii) $F = T (= mg)$

$$R = 2mg$$

B1

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B1

$$F \leq \mu R, \quad mg \leq 2mg\mu, \quad \mu \geq \frac{1}{2}$$

*Inequality needed, or convincing verbal explanation,
 ft if μ numeric and positive*

B1F

3

(b) $T = 2ma$

M1 either equation

M1A1

$$mg - T = ma$$

Accept $g = 9.8$ or 10 for either A1

A1

$$2mg - 2T = T$$

Full method for T

m1

$$T = \frac{2mg}{3}$$

ft one slip

If friction included, can either earn M1A1 for Q ,
A0 for P , m1A0.

SC Whole string method can earn m1 for $mg = 3ma$
and A1F for reult for T if M1 already earned

A1F

5

(accept: $T = 6.53m$ if 9.8 used, $T = 6.67m$ if 10 used,
 $T = 6.54m$ if 9.8 used and a found first)

[9]

Q14.

- (a) (i) trailer: $\pm S + 100 = 250 \times 0.5$
M1 for 3 terms, A1 all correct, allow $\pm S$
 g included with mass $\Rightarrow$ M1A0

M1A1

$$S = 25N$$

ft one error (including g present) provided $S > 0$
If S negative, accept sign changed for answer

A1F

3

- (ii) **car + trailer:**

M1

$$500 + 100 + R = 150 \times 0.5$$

All terms for M1, including appropriate mass
 g included with mass $\Rightarrow$ M1 A0 A1

or: car:

$$R + 500 - S = 1250 \times 0.5$$

–1 each error; ft S if used

A2, 1, 0

$$R = 150 \text{ N}$$

ft one slip, for $R > 0$

If R consistently in wrong direction, need explanatory comment re direction for A1, otherwise M1 A1 A0 A0 ($R < 0$)

A1F

4

(b) $T = 100 \text{ N}$

CAO

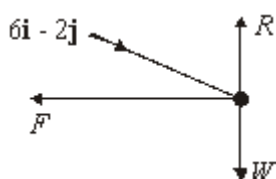
B1

1

[8]

Q15.

(a)



B1

1

(b) $6 - F = 1.5 \times 0.6$

3 terms for M1

M1 A1

$$F = 5.1 \text{ N}$$

CAO

A1

3

(c) $R = 2 + 1.5 \times 9.8$

RHS, 2 forces for M1

M1A1F

$$R = 16.7 \text{ N}$$

A1F

3

(d) $F = \mu R$
 $5.1 = \mu \times 16.7$

M1

$\mu = 0.305$

A1F

2

[9]

Q16.

(a) A: $0.21g - T = 0.21a$
'whole string' method 3/5

M1A1

B: $T - 0.14g = 0.14a$
M1A1 for equation, A1 for answer

A1

$0.07g = 0.35a$

M1

$a = 1.96 \text{ ms}^{-2}$
 CAO

A1

5

(b) $v = 0 + 1.96 \times 2$

M1

$v = 3.92 \text{ ms}^{-1}$

A1

2

(c) $0 = 3.92 - 9.8t$

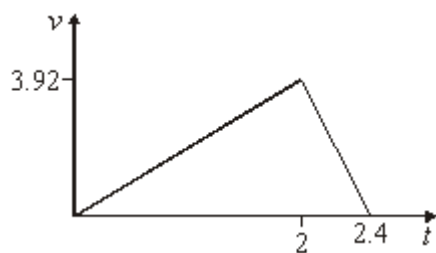
M1

$t = 0.4 \text{ sec}$

A1F

2

(d)



Two straight lines

B1

Appropriate gradients

B1

Labelling

B1

3

[12]

Q17.

(a) (i) $80 = \frac{1}{2} (0 + 20) t$
Use of constant acceleration equation to form an equation for t.

M1

$t = 8$ seconds
Correct time

A1

2

(ii) $20^2 = 0^2 + 2 \times a \times 80$
Use of constant acceleration equation to form an equation for a.

M1

$a = \frac{20^2}{160} = 2.5 \text{ ms}^{-2}$
ag *Correct acceleration from correct working.*

A1

2

(b) (i) $F = 1200 \times 2.5 = 3000 \text{ N}$
Use of $F = ma$

M1

Correct force.

A1

2

(ii) $F - 400 = 3000$

Three term equation of motion.

M1

$$F = 3400 \text{ N}$$

Correct F .

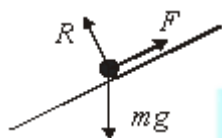
A1

2

[8]

Q18.

(a) (i)



Correct force diagram

B1

1

(ii) $R = 20 \times 9.8 \cos 40^\circ = 150 \text{ N}$

Resolving perpendicular to the slope

M1

ag Correct answer from correct working

A1

2

(iii) $F = 20 \times 9.8 \sin 40^\circ = 126 \text{ N}$

Resolving parallel to the slope

M1

$$126 = \mu \times 150$$

Correct friction

A1

$$\mu = \frac{126}{150} = 0.84$$

Use of $F = \mu R$

M1

ag Correct μ from correct working

A1

4

(b) It would be reduced

Correct explanation

B1

1

[8]

Q19.

(a) $-16\mathbf{i} + 16\mathbf{j} = \frac{1}{2} \mathbf{a} \times 4^2$

Use of vector constant acceleration equation

M1

Correct equation

A1

$$\mathbf{a} = \frac{-16\mathbf{i} + 16\mathbf{j}}{8} = -2\mathbf{i} + 2\mathbf{j}$$

Solving for $\mathbf{a}$

M1

ag Correct $\mathbf{a}$

A1

4

(b) $\mathbf{v} = 5(-2\mathbf{i} + 2\mathbf{j}) = -10\mathbf{i} + 10\mathbf{j}$

Expression for $\mathbf{v}$ at $t = 5$

M1

Correct expression

A1

$$v = \sqrt{10^2 + 10^2} = 14.1 \text{ ms}^{-1} \text{ (to 3 sf)}$$

Finding magnitude

M1

Correct speed

A1

4

(c) $20\mathbf{i} - 10\mathbf{j} + \mathbf{Q} = 3(-2\mathbf{i} + 2\mathbf{j})$

Application of Newton's second law to form equation

M1

Correct equation

A1

$\mathbf{Q} = (-20 - 6)\mathbf{i} + (10 + 6)\mathbf{j} = -26\mathbf{i} + 16\mathbf{j}$

Solving for Q

M1

Correct Q

A1

4

[12]

Q20.

(a) $150g - T = 150a$

Three term equation of motion for tank.

M1

$T - 200g \sin 30^\circ = 200a$

Correct equation

A1

$150g - 200a - 200g \sin 30^\circ = 150a$

Three term equation of motion for trolley

M1

Correct equation

A1

$$a = \frac{150g - 200g \sin 30^\circ}{350} = 1.4 \text{ ms}^{-2}$$

Solving for a

m1

Correct equation for a

A1

ag *Correct a from correct working.*

A1

7

(b) $T = mg$

Equations of motion for both bodies.

M1

$$T = 200 g \sin 30^\circ$$

Correct equation for tank.

A1

Correct equation for trolley

A1

$$m = 200 \sin 30^\circ = 100 \text{ kg}$$

Correct mass

A1

4

[11]

Q21.

(a) $4^2 = 20^2 + 2 \times 48a$

Use of constant acceleration equation to find a

M1

$$a = \frac{16 - 400}{96} = -4 \text{ ms}^{-2}$$

ag Correct acceleration from correct working

A1

2

(b) $0^2 = 4^2 + 2 \times (-4)s$

use of constant acceleration equation to form equation for s

M1

$$s = \frac{16}{8} = 2 \text{ m}$$

Correct equation

A1

Correct s

A1

3

(c) $0 = 20 - 4t$

Use of constant acceleration equation to form equation for t

M1

$$t = 5$$

Correct t

A1

2

(d) $-F - 200 = 1100 \times (-4)$

Three term equation of motion

M1

Correct equation

A1

$$F = 4200$$

Correct F

A1

3

[10]

Q22.

(a) $R = 2 \times 9.8 \cos 40^\circ$

Resolving perpendicular to the slope

M1

$$F = 0.3 \times 2 \times 9.8 \cos 40^\circ$$

Use of $F = \mu R$

M1

$$= 4.50 \text{ N}$$

Correct F

A1

3

(b) $2a = -2 \times 9.8 \sin 40^\circ - 4.50$

Three term equation of motion

M1

$$a = \frac{-2 \times 9.8 \sin 40^\circ - 4.50}{2} = -8.55 \text{ ms}^{-2}$$

Correct acceleration

A1

3

[6]

Q23.

(a) $T - 800 \times 9.8 = 800 \times 0.2$

Three term equation of motion

M1

Correct equation

A1

$T = 7840 + 160 = 8000 \text{ N}$

ag *Correct tension*

A1

3

(b) $T = 800 \times 9.8 = 800 \times (0.2)$

Three term equation of motion

M1

$T = 7840 - 160 = 7680 \text{ N}$

Correct tension

A1

2

(c) $T = 800 \times 9.8 = 7840 \text{ N}$

Correct tension

B1

1

[6]

Q24.

(a) $R = 20 \times 9.8 = 196 \text{ N}$

Finding R and using $F = \mu R$

M1

$F = 0.15 \times 196 = 29.4 \text{ N}$

Correct friction

A1

2

(b) $T - 29.4 = 20a$

Equation of motion for A

M1

Correct equation

A1

$$5 \times 9.8 - T = 5a$$

Equation of motion for B

M1

Correct equation

A1

$$49 - (20a + 29.4) = 5a$$

$$a = \frac{19.6}{25} = 0.784 \text{ ms}^{-2}$$

ag *Correct acceleration from correct working*

A1

5

(c) $T = 20 \times 0.784 + 29.4 = 45.1 \text{ N}$

Substituting a into one equation of motion

M1

Correct tension

A1

2

(d) $s = \frac{1}{2} \times 0.784 \times 3^2$

Use of constant acceleration equation with $u = 0$

M1

$$= 3.53 \text{ m}$$

Correct equation

A1

Correct distance

A1

3

[12]

Q25.

(a) (i) $6^2 = 2^2 + 2 \times a \times 10$

Use of a constant acceleration equation to find a

M1

$$a = \frac{36 - 4}{20} = 1.6 \text{ ms}^{-2}$$

Correct result from correct working

A1

2

(ii) $6 = 2 + 1.6t$

Use of a constant acceleration equation to find t

M1

$$t = \frac{4}{1.6} = 2.5 \text{ s}$$

Correct t from correct working

A1

2

(b) $F - 35 = 65 \times 1.6$

$$F = 104 + 35 = 139 \text{ N}$$

Three term equation of motion

M1

Correct equation

A1

Correct force

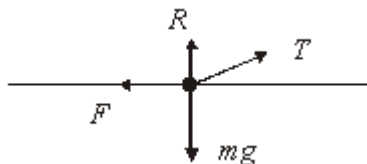
A1

3

[7]

Q26.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 40^\circ = 50 \times 9.8$

$R = 490 - T \sin 40^\circ$

Three term equation of motion

M1

Correct equation

A1

Correct expression for R

Use of $F R = \mu$

A1

3

(c) $F = 0.6 (490 - T \sin 40^\circ)$

Use of $F = \mu R$

M1

$= 294 - 0.6 T \sin 40^\circ$

ag *Correct result from correct working*

A1

2

(d) $T \cos 40^\circ - (294 - 0.6 T \sin 40^\circ) = 50 \times 0.5$

$T = \frac{319}{\cos 40^\circ + 0.6 \sin 40^\circ} = 277 \text{ N}$

Four term equation of motion

M1

Correct equation

A1

Solving for T

M1

Correct T

A1

4

[10]

Q27.

(a) $9 = 3 + 1.2t$

Forming constant acceleration equation

M1

$$t = \frac{9 - 3}{1.2} = 5 \text{ seconds}$$

Correct result from correct working

A1

2

(b) $s = \frac{1}{2} (3 + 9) \times 5 = 30 \text{ metres}$

Forming constant acceleration equation

Correct distance

M1 A1

2

(c) $F = 1200 \times 1.2 = 1440 \text{ N}$

Applying Newton's second law with $a = 1.2$

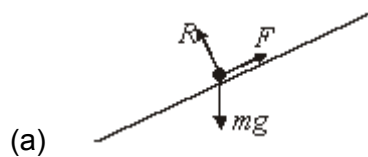
Correct F

M1 A1

2

[6]

Q28.



B1

1

(b) $R = 5 \times 9.8 \cos 40^\circ = 37.5 \text{ N}$

Correct force diagram

Resolving perpendicular to slope

Correct R

M1 A1

2

(c) $F = 0.2R = 7.51 \text{ N}$

Using $F = \mu R$

Correct F from correct working

M1 A1

2

(d) $5 \times 9.8 \sin 40^\circ - F = 5a$

Resolving parallel to slope to give 3 term equation of motion

M1 A1

Correct equation

$$a = \frac{5 \times 9.8 \sin 40^\circ - F}{5} = 4.80 \text{ ms}^{-2}$$

m1

Solving for a

Correct a from correct working

A1

4

(e) $10^2 = 2^2 + 2 \times 4.80s$

Forming constant acceleration equation

Correct equation

M1 A1

$$s = \frac{100 - 4}{9.6} = 10.0 \text{ m}$$

Correct s

A1

3

[12]

Q29.

(a) $14a = 14g \sin 45^\circ - T$

Equation of motion for one particle

M1

$$6a = T - 6g$$

Correct equation

A1

$$14a = 14g \sin 45^\circ - (6a + 6g)$$

Equation of motion for other particle

Correct equation

M1 A1

$$a = \frac{14g \sin 45^\circ - 6g}{20} = 1.91 \text{ ms}^{-2}$$

Correct a from correct working

M1 A1

6

(b) $T = mg$

Equation for one particle

M1

$T = 14g \cos 45^\circ$

*Equation of other particle
correct m*

M1 A1

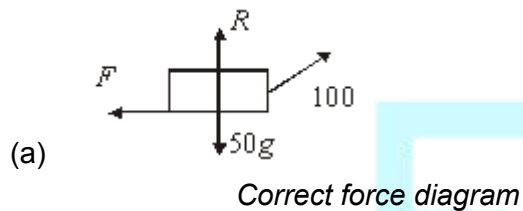
$m = 14 \cos 45^\circ = 9.90 \text{ kg}$

A1

4

[10]

Q30.



B1

1

(b) $50 \times 9.8 = R + 100 \sin 30^\circ$

*Resolving vertically
Correct equation*

M1 A1

$R = 440 \text{ N}$

Correct F from correct working

A1

3

(c) $100 \cos 30^\circ \leq \mu \times 440$

Use of $F \leq \mu R$ or $F = \mu R$

M1

$\therefore k = 0.197$

A1

$\mu \geq 0.197$

Correct equation

Correct k from correct working

A1

(d) $50a = 100 \cos 30^\circ - 0.1 \times 440$

Resolving horizontally to obtain a 3 term equation of motion

M1

3

Correct equation

A1

Solving for a

m1

$a = 0.852 \text{ ms}^{-2}$

Correct a

Allow 0.680 or 0.681

A1

4

[11]

Q31.

(a) (i) $s_1 = \frac{1}{2} \times 0.2 \times 8^2 = 6.4 \text{ m}$

Constant acceleration equation with $u = 0$

Correct distance

M1A1

2

(ii) $v = 0.2 \times 8 = 1.6 \text{ ms}^{-1}$

Constant acceleration equation with $u = 0$

M1

2

(iii) $s_2 = 1.6 \times 3 = 4.8 \text{ m}$

M1

$s = 6.4 + 4.8 + 1 = 12.2 \text{ m}$

Finding stage II distance

Correct total distance

A1

2

(b) $1 = \frac{1}{2} (0 + 1.6)t_3$
Equation to find time

M1

$t_3 = 1.25$
Correct time

A1

$t = 8 + 3 + 1.25 = 12.25 \text{ s}$
Correct total time

A1

3

(c) $T - 600 \times 9.8 = 600 \times 0.2$
Three term equation of motion
Correct equation

M1 A1

$T = 6000 \text{ N}$
Correct tension

A1

3

[12]

Q32.

(a) $R + 100 \sin 20^\circ + 60 \sin 50^\circ = 490$
Resolving vertically
Correct equation

M1 A1

$R = 410$

A1

3

(b) (i) $100 \cos 20^\circ - 60 \cos 50^\circ - F = 25$
Four term equation of motion
Correct equation

M1 A1

$F = 30.4$
Solving for F

Correct F from from correct working

M1 A1

4

(ii) $30.40 = \mu \times 409.8$

Use of $F = \mu R$

M1

$\mu = 0.0742$

Correct μ

A1

2

[9]

Q33.

(a) $40\mathbf{i} - 15\mathbf{j} = \frac{1}{2} (5\mathbf{i} - 2\mathbf{j} + \mathbf{v}) \times 10$

Use of constant acceleration equation with $\mathbf{v}$ unknown

Correct equation

M1 A1

$\mathbf{v} = 8\mathbf{i} - 3\mathbf{j} - 5\mathbf{i} + 2\mathbf{j} = 3\mathbf{i} - \mathbf{j}$

Solving for $\mathbf{v}$

Correct $\mathbf{v}$

M1 A1

4

(b) $3\mathbf{i} - \mathbf{j} = 5\mathbf{i} - 2\mathbf{j} + 10\mathbf{a}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1A1

$\mathbf{a} = -0.2\mathbf{i} + 0.1\mathbf{j}$

Correct $\mathbf{a}$

A1

3

(c) $\mathbf{F} = 15(-0.2\mathbf{i} + 0.1\mathbf{j})$

Use of $\mathbf{F} = m\mathbf{a}$

M1

$= -3\mathbf{i} + 1.5\mathbf{j}$

Correct $\mathbf{F}$

A1

$$F = \sqrt{3^2 + 1.5^2} = 3.35$$

Finding magnitude of F

Correct magnitude

M1 A1

4

[11]

Q34.

(a) $R = 20 \times 9.8 = 196$

cao

B1

1

(b) $F \leq 0.3 \times 196 = 58.8$

Using 0.3×196

M1

If $P = 80$, $F = 58.8$

58.8 as answer

A1

If $P = 40$, $F = 40$

40 as answer

A1

3

(c) $P - 58.8 = 20 \times 0.8$ © Exam Papers Practice. All rights reserved.

Three term equation of motion including 58.8

M1

Correct equation

A1

$P = 74.8$

A1

Correct P

A1

3

(d) $a = \frac{-58.8}{20} = -2.94$

Use of $F = ma$ with ± 58.8

M1

Correct acceleration with a negative sign

A1

$$0^2 = 6^2 + 2 \times (-2.94)s$$

Use of $v^2 = u^2 + 2as$ with $v = 0$

m1 A1

$$s = \frac{36}{5.88} = 6.12$$

Correct distance

A1

5

[12]

Q35.

(a) $0.5 = \frac{1}{2} \times a \times 4$

Use of $s = ut + \frac{1}{2}at^2$ with $u = 0$

M1

$$a = 0.25$$

Correct acceleration

A1

2

(b) $6 \times 9.8 - T = 6 \times 0.25$

*Three term equation of motion for particle,
with correct use of g*

M1

Correct equation

A1ft

$$T = 57.3$$

ag *Correct T from correct working*

A1ft

3

(c) $57.3 - F = 10 \times 0.25$

Three term equation of motion for the block

M1

Correct equation

A1ft

$F = 54.8$

Correct F

A1ft

$R = 10 \times 9.8 = 98$

R = 98 seen in working

B1

$54.8 = 98\mu$

Use of $F = \mu R$

m1

$$\mu = \frac{54.8}{98} = 0.559$$
correct μ

A1ft

6

[11]